MOVZZ — Master Roadmap (Combined)

> Reliability-Orchestrated Mobility Platform · Chennai, India Last updated: March 11, 2026

Legend

Symbol	Meaning
☑ DONE	Implemented and working
🔴 CRITICAL	Security risk / launch blocker
🟠 HIGH	Needed for beta users
🟡 MEDIUM	Important but deferrable post-launch
🟢 NICE	Post-launch enhancement
☐ TODO	Not yet started
🤖 AI	AI engine task

Grand Summary

Category	Total	Done	Remaining
Section 0 — Security Hardening	11	11	0
Section 1 — Foundation	34	34	0
Section 2 — Platform Features	26	11	15
Section 3 — AI Intelligence Layer	23	18	5
Section 4 — Production & Deployment	14	0	14
TOTAL	**108**	**74**	**34**

Progress: 74/108 tasks (69%)

SECTION 0 — Security Hardening ☑ (11/11 Complete)

> Completed March 10, 2026 — all 11 vulnerabilities patched.

#	Priority	Task	Status
S1	🔴	**Remove hardcoded JWT fallback secret** — throws `Error('JWT_SECRET is required')` on startup	☑ DONE
S2	🔴	**Fix auth race condition** — `auth.controller.ts` uses `prisma.user.upsert()` — atomic, no duplicate crash	☑ DONE
S3	🔴	**Fix OAuth race condition** — `oauth.controller.ts` uses `upsert` — safe under concurrent logins	☑ DONE
S4	🔴	**Add admin role protection** — `role String @default("user")` in schema; `requireAdmin` middleware on all `/admin/` routes	☑ DONE

#	Priority	Task	Status
S5	⬤	**Fix hardcoded API URL** — all three files now use `VITE_API_URL` env var with localhost fallback	☑ DONE
S6	⬤	**Protect env files** — `.gitignore` fixed; `backend/.env.example` + `frontend/.env.example` created	☑ DONE
S7	⬤	**OTP rate limiting** — `express-rate-limit` on `/verify-otp`: 5 attempts per phone per 10 min	☑ DONE
S8	⬤	**Fix Socket memory leak** — handler refs stored; `socket.off()` called with same refs on disconnect	☑ DONE
S9	⬤	**Fix dual fare engine** — `quotes.service.ts` FARE_CONFIGS aligned with `fare.service.ts` rates	☑ DONE
S10	⬤	**Add token expiry** — `tokenExpiry = Date.now() + 7d` on login; auto-logout on startup if expired	☑ DONE
S11	⬤	**Harden CORS + global rate limit** — CORS locked to allowlist; 100 req/15 min global limiter	☑ DONE

SECTION 1 — Foundation ☑ (34/34 Complete)

Phase 1 — Core Backend ☑ (12/12)

#	Task	Status	Key File(s)
1	Express + TypeScript + Prisma scaffold	☑ DONE	`backend/src/index.ts`
2	PostgreSQL schema — Users, Providers, Bookings, BookingAttempts, BookingLogs	☑ DONE	`backend/prisma/schema.prisma`
3	Redis cache with in-memory fallback	☑ DONE	`backend/src/config/redis.ts`
4	OTP authentication (phone) with JWT	☑ DONE	`backend/src/controllers/auth.controller.ts`
5	Fare engine — mode-specific rates, surge, Haversine×1.35, airport detection, paise precision	☑ DONE	`backend/src/services/fare.service.ts`
6	Provider scoring — reliability from historical success rate	☑ DONE	`backend/src/services/provider-scoring.service.ts`
7	Booking state machine — SEARCHING → CONFIRMED → IN_PROGRESS → COMPLETED / FAILED / CANCELLED / MANUAL_ESCALATION	☑ DONE	`backend/src/services/booking.service.ts`
8	Recovery service — 3-level retry, auto-escalation, ₹100 compensation credit	☑ DONE	`backend/src/services/recovery.service.ts`
9	Quotes API — ranked options with reliability rationale	☑ DONE	`backend/src/services/quotes.service.ts`

#	Task	Status	Key File(s)
10	Booking API — create, poll status, cancel	☑ DONE	`backend/src/controllers/booking.controller.ts`
11	Admin dashboard API — stats, provider list	☑ DONE	`backend/src/controllers/admin.controller.ts`
12	Rate limiting, CORS, Helmet security headers	☑ DONE	`backend/src/index.ts`

Phase 2 — Frontend Prototype ☑ (7/7)

#	Task	Status	Key File(s)
13	React + Vite scaffold with Zustand state management	☑ DONE	`frontend/src/App.jsx`
14	5-screen flow: Landing → Auth → Transport → Destination → Results	☑ DONE	`frontend/src/App.jsx`
15	Transport mode selector — CAB (Economy/Comfort/Premium), Bike, Auto, Metro	☑ DONE	`frontend/src/App.jsx`
16	Results screen — ranked quote cards with reliability score, ETA, price, tags	☑ DONE	`frontend/src/App.jsx`
17	Auth store — OTP send/verify, JWT storage, auth state	☑ DONE	`frontend/src/stores/authStore.ts`
18	Booking store — fetchQuotes, createBooking, Socket.IO state sync	☑ DONE	`frontend/src/stores/bookingStore.ts`
19	API client (axios) with JWT header injection	☑ DONE	`frontend/src/api/client.ts`

Phase 3 — Production Hardening ☑ (5/5)

#	Task	Status	Notes
20	WebSockets (Socket.IO) — `booking:state_changed` server-push, per-user rooms, JWT on handshake	☑ DONE	`backend/src/config/socket.ts`
21	Google OAuth2 — direct REST flow, no Passport.js	☑ DONE	`backend/src/controllers/oauth.controller.ts`
22	Mapbox geocoding autocomplete — 300ms debounce, India filter	☑ DONE	`frontend/src/App.jsx`
23	Interactive map — react-map-gl, green/orange markers, auto-pan	☑ DONE	`frontend/src/App.jsx`
24	Chennai preset location chips with real lat/lng	☑ DONE	`frontend/src/App.jsx`

Phase 4 — Background Jobs / BullMQ ☑ (4/4)

#	Task	Status	Notes

#	Task	Status	Notes
25	BullMQ queue singletons — booking-timeout, recovery-retry, sms-dispatch, nightly-aggregation, ml-data-collection	☑ DONE	`backend/src/config/queues.ts`
26	Booking timeout worker — auto-cancel SEARCHING after 5 min	☑ DONE	`backend/src/workers/booking-timeout.worker.ts`
27	Recovery retry worker — async provider retry, 2s delay	☑ DONE	`backend/src/workers/recovery.worker.ts`
28	SMS dispatch worker — 3 retries, exponential backoff, Twilio-ready (mock in dev)	☑ DONE	`backend/src/workers/sms.worker.ts`

Phase 5 — Payments ☑ (5/5)

#	Task	Status	Notes
29	Razorpay — create payment link endpoint	☑ DONE	`POST /api/v1/payments/create-link`
30	Razorpay — verify payment + HMAC-SHA256 signature check	☑ DONE	`POST /api/v1/payments/verify`
31	Frontend — Razorpay Payment Link redirect + return detection	☑ DONE	`frontend/src/App.jsx`
32	Provider payout tracking (T+2 terms)	☑ DONE	`Payout` table, `payment.service.ts`
NEW	**Razorpay webhook handler** — server-to-server `payment_link.paid` event; HMAC-SHA256 verification; idempotent; auto-schedules T+2 payout	☑ DONE	`POST /api/v1/payments/webhook`, `payment.service.ts`

Phase 9 — Admin Panel ☑ (5/5)

#	Task	Status	Notes
33	Full Admin Panel UI at `/admin` — 4 tabs: Dashboard, Escalations, Providers, Metrics	☑ DONE	`frontend/src/Admin.jsx`
34	Live booking map — react-map-gl, color-coded markers by state	☑ DONE	`frontend/src/Admin.jsx`
35	Provider management — list, pause/resume, add new provider	☑ DONE	`frontend/src/Admin.jsx`
36	Manual escalation queue — ops confirms MANUAL_ESCALATION bookings	☑ DONE	`frontend/src/Admin.jsx`
37	Analytics tab — today's booking states, weekly revenue, top providers	☑ DONE	`frontend/src/Admin.jsx`

Phase 7 — Infrastructure (partial) ☑ (3/4)

#	Task	Status	Notes

#	Task	Status	Notes
38	Sentry error tracking — `@sentry/node` + `@sentry/react`, gated on env var	☑ DONE	`backend/src/index.ts`, `frontend/src/main.jsx`
39	Neon.tech — cloud Postgres DATABASE_URL swap ready in `backend/.env`	☑ DONE	`backend/.env` (commented format)
40	AWS S3 — presigned URL pattern, profile photo upload endpoint	☑ DONE	`backend/src/services/s3.service.ts`

Phase 6 — Notifications (partial) ☑ (1/4)

#	Task	Status	Notes
41	Transactional email via Resend — confirmation, cancellation, compensation	☑ DONE	`backend/src/services/email.service.ts`

Phase 10 — Mobile (partial) ☑ (1/6)

#	Task	Status	Notes
42	PWA — vite-plugin-pwa, service worker, web manifest, offline cache, installable	☑ DONE	`frontend/vite.config.js`

SECTION 2 — Platform Features 🔄 (11/26 Done, 15 Remaining)

Phase 6 — Notifications (remaining) 🔄 (3/4 done)

#	Priority	Task	Notes
43	🟤	**WhatsApp OTP delivery** — `sms.worker.ts` now sends via Twilio WhatsApp API; dev sandbox + prod path; console fallback if creds absent	☑ DONE — `backend/src/workers/sms.worker.ts`
44	🟤	FCM push notifications — booking state changes, driver ETA	Needs React Native or PWA service worker + Firebase project
45	🟢	WhatsApp booking alerts + driver confirmation	OTP via WhatsApp ☑ DONE. Booking state alerts (post-launch) — same Twilio creds, extend sms.worker.ts

Phase 7 — Infrastructure (remaining) 🔄 (3/4 done)

#	Priority	Task	Notes
46	🟢	CDN — CloudFront or Cloudflare in front of S3	Quick win ~1 hr after S3 verified

Phase 8 — Provider Integrations ☐ (0/6 — all new)

#	Priority	Task	File(s)

#	Priority	Task	File(s)
P0	⬤	**Create `IProvider` interface + `ProviderRegistry`** — standard contract for all providers: `getEstimate()`, `checkAvailability()`, `bookRide?()`	`backend/src/services/providers/provider.interface.ts` NEW, `provider.registry.ts` NEW
47	⬤	**Fast Track Cabs integration** — WhatsApp Business API booking or manual confirmation flow; show price with "confirmation required" tag	`backend/src/services/providers/fasttrack.provider.ts` NEW
48	⬤	**Chennai Call Taxi integration** — same pattern as Fast Track	`backend/src/services/providers/chennai-call-taxi.provider.ts` NEW
49	⬤	Uber/Ola/Rapido aggregator fallback — reverse-engineer price estimate APIs (no booking); Redis cache 5 min; graceful fallback if unavailable	`uber.provider.ts`, `ola.provider.ts`, `rapido.provider.ts` NEW
50	⬤	Provider onboarding portal — KYC + doc upload UI; `s3.providerDocKey()` already wired	Admin UI new tab + upload flow
P1	⬤	**Provider Health Dashboard** — new Admin tab: status, success rate, response time, rate limits per provider	`frontend/src/Admin.jsx`, `GET /api/v1/admin/providers/health` NEW

Phase 10 — Mobile Apps (remaining) 🔄 (1/6 done)

#	Priority	Task	Notes
51	⬤	React Native user app — port 5-screen flow (Expo recommended)	Reuse existing API + WebSocket
52	⬤	React Native driver app — accept/reject rides, navigation	Separate app; driver-specific Socket.IO events
53	⬤	Driver location tracking — real-time GPS `driver:location_update` event	New event in `backend/src/config/socket.ts`
54	⬤	In-app FCM push notifications	Firebase + React Native Push Notification library
55	⬤	App Store + Play Store deployment	Expo EAS build

SECTION 3 — AI Intelligence Layer 🤖 (18/23 Done, 5 Remaining)

> **Achieved targets (post AI Week 1):** Booking success rate 75% → tracked. Confirmation time 45s → tracked.
> **Remaining target:** Revenue +10–15% via fair dynamic pricing (AI Week 3).

AI Week 1 — Core AI Foundation ☑ (8/8)

#	Task	Status	Files
AI-1	TypeScript AI interfaces — `RideContext`, `ProviderMetrics`, `ReliabilityScore`, `ScoreBreakdown`, all enums	☑ DONE	`backend/src/types/ai.types.ts`
AI-2	Context Builder Service — Chennai zone detection, Haversine distance, peak hours, weekend, ride type	☑ DONE	`backend/src/services/context-builder.service.ts`
AI-3	Provider Metrics Service — hourly rates, recency windows (1h/6h/24h/7d), streaks, Redis 5-min cache	☑ DONE	`backend/src/services/provider-metrics.service.ts`
AI-4	Reliability Predictor — `predictReliability()` with 8 contextual adjustments	☑ DONE	`backend/src/services/provider-scoring.service.ts`
AI-5	Orchestration Service — `decideStrategy()` + `executeStrategy()` with SEQUENTIAL / PARALLEL_2 / PARALLEL_3 / CASCADE / EMERGENCY	☑ DONE	`backend/src/services/ai/orchestration.service.ts`
AI-6	Failure Detector Service — risk scoring, tiered interventions, ₹50 credit at riskScore ≥ 85	☑ DONE	`backend/src/services/ai/failure-detector.service.ts`
AI-7	Prisma migration — AI fields on Booking + Provider	☑ DONE	`backend/prisma/schema.prisma`
AI-8	Wire AI into booking.service.ts — `assignWithAI()` with fallback to legacy `assignProvider()`	☑ DONE	`backend/src/services/booking.service.ts`

AI Week 2 — Performance + Visible AI in UI ☑ (5/5)

#	Priority	Task	Status	Files
AI-9	⚫	**Nightly Aggregation Worker** — BullMQ CRON at midnight; per-provider success rates; rolling reliability (90%+10%); invalidates Redis cache	☑ DONE	`backend/src/workers/nightly-aggregation.worker.ts`
AI-10	⚫	**Cache Service** — `getOrCompute(key, ttl, fn)` pattern; `CacheKeys` builder; `TTL` constants; `invalidateProviderScores()`	☑ DONE	`backend/src/services/ai/cache.service.ts`

#	Priority	Task	Status	Files
AI-11	⚪	**ML Data Collection Worker** — triggers on COMPLETED/FAILED/CANCELLED; writes `MLTrainingData` row with context, scores, outcome, distance, hour, day	☑ DONE	`backend/src/workers/ml-data.worker.ts`
AI-12	⚪	**Prisma schema — AI Week 2 tables** — `ProviderMetricsCache` (Redis-backed aggregated metrics) + `MLTrainingData` (booking outcome training rows)	☑ DONE	`backend/prisma/schema.prisma`
AI-13	⚪	**Frontend AI-enhanced results screen** — reliability progress bar (green/yellow/red), AI reasoning text (`why`), emoji badges, MOVZZ score chip, animated booking status with orchestration strategy badge	☑ DONE	`frontend/src/App.jsx`

AI Week 3 — Demand Forecasting + Fair Dynamic Pricing ☑ (5/5)

> Completed March 11, 2026 — all 5 tasks done.

#	Priority	Task	Status	Files
AI-14	⚪	**Demand Forecaster Service** — 4-week rolling avg per zone/hour; LOW/NORMAL/HIGH/VERY_HIGH levels (×1.0/1.0/1.08/1.15)	☑ DONE	`backend/src/services/ai/demand-forecaster.service.ts`
AI-15	⚪	**Prisma migration — demand forecasts** — `DemandForecast` table (zone+forecastHour unique, predictedRides, confidence, actualRides, forecastAccuracy); migration SQL created manually (DB offline)	☑ DONE	`backend/prisma/schema.prisma`
AI-16	⚪	**Demand forecast worker** — BullMQ CRON at 10 PM nightly; 7 zones × 24h forecasts; upserts `DemandForecast`; SMS nudge to drivers in shortage zones with ₹100 incentive	☑ DONE	`backend/src/workers/demand-forecast.worker.ts`

#	Priority	Task	Status	Files
AI-17	⚪	**Enhanced dynamic pricing** — `getDynamicMultiplier()` in fare.service.ts; weather + traffic + demand multipliers; **hard cap: ×1.2**; returns `{ finalMultiplier, breakdown[], explanation }`	☑ DONE	`backend/src/services/fare.service.ts`
AI-18	⚪	**Pricing breakdown API + frontend** — `quotes.controller.ts` attaches `baseFare`, `breakdown[]`, `explanation` per quote; App.jsx "Why this price? ▼" expandable panel per card; only visible when breakdown non-empty	☑ DONE	`backend/src/controllers/quotes.controller.ts`, `frontend/src/App.jsx`

AI Week 4 — Provider Analytics + User Personalization ☐ (0/5)

#	Priority	Task	Files
AI-19	⚪	**Provider Analytics Service** — behavioral profiles: airport specialist, morning vs night, price-sensitive vs volume; human-readable recommendations	`backend/src/services/ai/provider-analytics.service.ts` NEW
AI-20	⚪	**Provider Dashboard Endpoint** — `GET /api/v1/providers/:id/insights` — overallStats, bestTimes, preferredZones, earningsOptimization, aiInsights	`backend/src/controllers/provider-dashboard.controller.ts` NEW
AI-21	⚪	**User Personalization Service** — favorite providers, commute detection, pre-book vs last-minute, price sensitivity; `GET /api/v1/users/me/recommendations`	`backend/src/services/ai/personalization.service.ts` NEW
AI-22	⚪	**Prisma migration — personalization tables** — `UserPreference` + `UserPattern`	`backend/prisma/schema.prisma`
AI-23	⚪	**Admin AI Dashboard** — orchestration strategy distribution, prediction accuracy, AI vs non-AI success rate, demand vs actual heatmap	`backend/src/controllers/admin.controller.ts`, `frontend/src/Admin.jsx`

SECTION 4 — Production & Deployment ☐ (0/14 — All New)

> Required before any real users. Run after Section 0 security fixes are complete.

Production Setup

#	Priority	Task	Notes
D1	●	**Railway project setup** — create Postgres + Redis + Node.js service; connect GitHub repo; auto-deploy on push	railway.app — ~₹3,000–5,000/month
D2	●	**Frontend deploy to Vercel** — import repo; set `VITE_API_URL` to Railway backend URL; SSL auto-configured	Auto-deploys on every git push to main
D3	●	**Domain setup** — buy movzz.in; point DNS to Railway/Vercel; SSL configured automatically	₹500–1,000/year
D4	●	**Production DB migration** — run `npx prisma migrate deploy` on Railway; seed initial data: Chennai zones, transport modes, base fare rates, provider list	Railway console
D5	●	**Set all production env vars** — JWT_SECRET (64-char random), Razorpay live keys, Twilio, Resend, Mapbox, Sentry DSNs, Google OAuth callback URL, RAZORPAY_WEBHOOK_SECRET	Railway dashboard env vars
D6	○	**Razorpay production activation** — switch test → live keys; complete KYC (2–3 business days); set webhook `https://movzz.in/api/v1/payments/webhook`; test ₹1 live transaction	Razorpay dashboard
D7	○	**Seed first admin user** — `UPDATE "User" SET role = 'admin' WHERE phone = '+91...'` via Prisma	Production DB

Testing

#	Priority	Task	Notes
D8	○	**Load testing (k6)** — ramp 10 → 50 → 100 concurrent users; target: 95% requests < 2s, 0% error, CPU < 70%	`backend/tests/load-test.js` NEW
D9	○	**End-to-end user testing** — 5–10 testers; flow: signup → OTP → quote → booking → SMS → status update	Manual test plan
D10	○	**Mobile responsiveness audit** — iPhone Safari + Android Chrome; min 44×44px touch targets; no horizontal scroll	Chrome DevTools + real devices

Legal & Documentation

#	Priority	Task	Notes
D11	○	**Terms of Service** — termly.io template; customize for ride-booking and reliability guarantee	`frontend/src/pages/Terms.jsx` NEW
D12	○	**Privacy Policy** — DPDP Act (India) compliance basics	`frontend/src/pages/Privacy.jsx` NEW
D13	○	**User FAQ / Help Center** — 4 sections: Getting Started, Pricing, Reliability, Account	`frontend/src/pages/Help.jsx` NEW
D14	○	**Env documentation** — fully document all env vars in both `.env.example` files; update README with local setup	`backend/.env.example`, `frontend/.env.example`, `README.md`

SECTION 5 — Execution Order Summary

```
COMPLETED ☑:
  S1–S11                        ← Security hardening (all 11 done)
  #1–#42                        ← Full foundation (all 33 + webhook done)
  AI-1 through AI-18            ← AI Weeks 1–3 (18/23 done)
  #43 #45 (partial)             ← WhatsApp OTP live (Spotify-style message format)
  Razorpay webhook              ← Server-to-server payment confirmation

NEXT (before beta):
  P0 #47 #48                    ← ⬤ Provider interface + first 2 real providers
  P1                            ← ⬤ Provider health dashboard
  D1 D2 D3 D4 D5               ← ⬤ Deploy to production
  D6 D7 D8 D9 D10              ← ⬤ Activate services + test
  D11 D12                       ← ⬤ Legal (Terms + Privacy)

POST-LAUNCH:
  #49 #50                       ← ⬤ Uber/Ola/Rapido + provider onboarding
  #45 #46                       ← ⬤ WhatsApp + CDN
  D13 D14                       ← ⬤ FAQ + env docs
  AI-19 AI-20 AI-21 AI-22 AI-23 ← ⬤ AI Week 4 (analytics + personalization)
  #44 #51 #52 #53 #54 #55      ← ⬤ FCM + React Native mobile apps
```

AI Success Metrics (Track Weekly After AI Week 1)

Metric	Baseline	Target (Post-AI)
Booking success rate	~75%	92%+
Avg time to confirmation	~45 seconds	<20 seconds
Rides needing manual ops	~20%	<5%
AI prediction accuracy	—	>90% (predicted vs actual outcome)
Revenue from dynamic pricing	0%	+10–15% (AI Week 3)
User satisfaction (avg rating)	~4.2	4.7+

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*MOVZZ Master Roadmap · March 13, 2026 · **74/108 tasks done (69%)***